Kernels and intersections in quotients of hypertopes

Human Compression project

Abstract

Injectivity on the maximal parabolic subgroups does not by itself preserve the intersection property under a quotient. We give an affine example in which every maximal parabolic embeds and the quotient geometry is flag transitive, but has only one chamber. We then state and prove the precise additional condition on products of parabolic subgroups. This corrects the quotient assertion in Theorem 3.2 of the fixed manuscript *An exploration of locally spherical regular hypertopes*. It does not dispose of that manuscript's classification results.

1 The direction of a quotient criterion

Let I be a finite set, and let $U = \langle s_i : i \in I \rangle$ be generated by involutions. Write $U_A = \langle s_i : i \in A \rangle$ for $A \subseteq I$ and $U_i = U_{I\setminus\{i\}}$. The marking is a C-group marking when

$$U_A \cap U_B = U_{A\cap B} \qquad (A, B \subseteq I).$$

All subgroups in this note refer to the specified marking.

The source's Theorem 3.1 is a valid upward criterion: if a marked epimorphism $G \longrightarrow H$ is injective on each maximal parabolic of G, and H is a C-group, then G is a C-group. Indeed, its proper parabolics inherit the property from their isomorphic images in H. An element of $G_i \cap G_j$ maps into $H_{I\setminus\{i,j\}}$; lift it within $G_{I\setminus\{i,j\}}$ and use injectivity in G_i. The maximal-parabolic criterion proved below completes the argument.

The proof of Theorem 3.2 instead uses a map $U/N \longrightarrow U$ which takes s_iN to s_i. Local injectivity does not make this map well defined: an additional relation in the quotient need not hold in U. The issue is not confined to that proof. The following example satisfies the stated hypotheses and contradicts its conclusion.

2 An affine example

Equip $\mathbb{R}^2$, in coordinates (a, b), with the inner product whose matrix is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Use homogeneous coordinates for the affine maps

$$s_0 = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad s_1 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

They are reflections in $a+b = 1$, $2a-b = 0$, and $-a+2b = 0$, respectively. Put $U = \langle s_0, s_1, s_2 \rangle$ and $W = \langle s_1, s_2 \rangle$. The latter is the group of six linear symmetries generated by the last two reflections. If t_v denotes translation by v, then

$$s_\theta = s_1 s_2 s_1 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad s_0 = t_{(1,1)} s_\theta.$$

Conjugating $t_{(1,1)}$ by s_1 gives $t_{(0,1)}$, and their quotient gives $t_{(1,0)}$. Every generator has integral translation part and linear part in W. Consequently

$$U = \mathbb{Z}^2 \rtimes W, \qquad N = \{t_v : v \in \mathbb{Z}^2\} \lhd U, \qquad U/N \cong W \cong S_3. \tag{1}$$

We verify that U is the full type-preserving group of a regular hypertope, so no converse about arbitrary marked groups is being assumed. The lines

$$2a - b \in \mathbb{Z}, \qquad -a + 2b \in \mathbb{Z}, \qquad a + b \in \mathbb{Z}$$

tile the plane by equilateral triangles. A fundamental triangle is

$$\Delta = \{2a - b \geq 0,\ -a + 2b \geq 0,\ a + b \leq 1\},$$

with vertices 0, $\lambda_1 = (2/3, 1/3)$, and $\lambda_2 = (1/3, 2/3)$. The vertex set is the disjoint union of $\mathbb{Z}^2$, $\lambda_1 + \mathbb{Z}^2$, and $\lambda_2 + \mathbb{Z}^2$. Give these three sets different types. Each generator preserves each set, as follows directly from the displayed matrices. Declare vertices incident when they coincide or are joined by a tiling edge.

Each edge belongs to two triangles; each vertex link is a hexagon; every three pairwise adjacent vertices form a tiling triangle. It follows that this incidence structure is a geometry, is thin, and is residually connected. Here thinness concerns extensions of an edge by a vertex of the third type, and residual connectivity concerns the whole tiling graph and its hexagonal vertex links.

The group U is transitive on triangles. To see this, cross a path of adjacent triangles starting at Δ. Once the current triangle is $u\Delta$, reflection in one of its sides is us_iu^{-1}; the next triangle is therefore $us_i\Delta$. The adjacency graph is connected, so induction reaches every triangle. The action is free on triangles. An element stabilizing Δ preserves types and hence fixes each of its three noncollinear vertices, which forces an affine map to be the identity. Finally, a type-preserving automorphism fixing one chamber fixes each neighboring chamber, since an edge has exactly two extensions. Propagation across the connected chamber graph fixes the whole geometry. Thus U is its full type-preserving automorphism group, acting regularly on chambers. Denote this regular hypertope by $\mathcal{U}$.

Proposition 1. *For each $i \in \{0, 1, 2\}$, the subgroup U_i has order six and $N \cap U_i = 1$. Nevertheless the coset geometry with group U/N and maximal marked parabolics $\pi(U_i)$ has one face of each type. It is flag transitive and is not a hypertope.*

Proof. Any pair of the three reflection matrices consists of involutions whose product has order three. A group generated by two such involutions is a quotient of the dihedral group of order six: every word reduces to $(s_is_j)^k$ or $(s_is_j)^ks_i$, $0 \leq k < 3$. The linear parts of each pair are distinct transpositions generating the six-element group W. Hence each pair generates exactly six elements and the linear-part map is injective on that subgroup. This proves $N \cap U_i = 1$ and $\pi(U_i) = W$ for all i.

Each coset set $W/\pi(U_i)$ is therefore a singleton. With types kept distinct, all three faces are incident and there is exactly one chamber. Every flag extends to it and W is transitive on flags of each prescribed type. Every residue of rank one has one face, so the geometry is not thin. The action of W has kernel W; faithfulness was not a hypothesis of the source theorem. The orbit quotient $\mathcal{U}/N$ is exactly the same geometry, since translations are transitive on vertices of each type. □

3 The additional kernel condition

Proposition 2. *Suppose $G = \langle r_i : i \in I\rangle$, with $|I| \geq 2$, and all maximal parabolics G_i are C-groups on their indicated generators. Then G is a C-group if and only if $G_i \cap G_j = G_{I\setminus\{i,j\}}$ for all distinct i, j.*

Proof. Necessity follows from the definition. For sufficiency, let $x \in G_A \cap G_B$. If either index set is I, the conclusion is immediate. Otherwise choose $i \notin A$ and $j \notin B$. If $i = j$, the desired intersection follows inside the C-group G_i. If $i \neq j$, the assumed maximal intersection places x in $G_{I\setminus\{i,j\}}$. Intersecting with G_A inside G_i places x in $G_{A\setminus\{j\}}$. Both this subgroup and G_B lie in G_j, whose intersection property now gives $x \in G_{A\cap B}$. The reverse inclusion always holds. □

Theorem 3. *Let $U = \langle s_i : i \in I\rangle$ be a C-group with $|I| \geq 2$, let $N \triangleleft U$, and assume $N \cap U_i = 1$ for every i. The induced marking on U/N is a C-group marking if and only if*

$$N \cap U_i U_j = \{1\} \qquad (i \neq j). \tag{2}$$

In particular, if (2) holds and the quotient coset geometry is flag transitive, it is a regular hypertope.

Proof. Write $\pi : U \longrightarrow U/N$ and $H_i = \pi(U_i)$. The local injectivity assumption identifies H_i, with its marking, with the C-group U_i.

Suppose (2) holds, and let $x \in H_i \cap H_j$. Choose $a \in U_i$ and $b \in U_j$ with $\pi(a) = x = \pi(b)$. Then $ab^{-1} \in N \cap U_i U_j$, so $a = b$. As U is a C-group, this common element lies in $U_i \cap U_j = U_{I\setminus\{i,j\}}$. Thus $H_i \cap H_j = \pi(U_{I\setminus\{i,j\}})$. Proposition 2 gives the full intersection property.

Conversely suppose U/N is a C-group. Write an element of $N \cap U_i U_j$ as ab^{-1} with $a \in U_i$, $b \in U_j$. The common image of a and b lies in $H_i \cap H_j = \pi(U_{I\setminus\{i,j\}})$. Choose a preimage c in that smaller subgroup. Injectivity on U_i gives $a = c$, and injectivity on U_j gives $b = c$. Hence $ab^{-1} = 1$.

For the final assertion, apply the C-group reconstruction theorem under its flag-transitivity hypothesis. This is Theorem 2.3 of the fixed source, whose complete proof is acquired from the project's reviewed manuscript on residue stabilizers and distinguished generators. □

The missing condition can be seen explicitly in our example: $s_\theta \in U_0$, $s_0 \in U_1$, but $s_\theta s_0 = t_{(-1,-1)}$ is a nonidentity element of $N \cap U_0 U_1$. Thus the repair excludes the example for the precise reason isolated in the proof.

Scope and verification. The fixed source is a sixteen-page revised author manuscript. The affected assertion and its proof are on page 6. Proposition 2 is its Proposition 2.1, attributed there to earlier work, and is reproved here for completeness. The affine group and its triangular complex are standard; the explicit argument above supplies all properties needed by the example. The kernel test is an elementary quotient-intersection criterion; no claim of historical priority is made. The accompanying program checks the displayed integer matrices, all three six-element parabolics, their exact linear images, and the nonzero kernel witness. The claims concerning the infinite tiling use the written proof. This note corrects one auxiliary theorem. It is neither a solution of an open conjecture nor a complete treatment of the source's spherical, Euclidean, and hyperbolic classifications.

References

[1] M. E. Fernandes, D. Leemans, and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*. Fixed sixteen-page revised author manuscript, catalogue L033, especially Propositions 2.1–2.2 and Theorems 2.3, 3.1–3.2.

[2] Human Compression project, *Residue stabilizers and distinguished generators*. Complete reviewed L052 general component; only the reconstruction theorem with its stated flag-transitivity hypothesis is used here.